

Prompt-to-Weights in the Small-Zoo Regime

A one-month hypernetwork project: generating LoRA adapters from data prompts on a single RTX 5090

Portfolio / research project plan • Target hardware: 1x RTX 5090 (32 GB, Blackwell) • Timeline: 4 weeks • Base method: Drag-and-Drop LLMs (DnD)

1 Objective

Train a hypernetwork that, conditioned on a batch of unlabeled task prompts, emits LoRA adapter weights for a frozen base LLM in a single forward pass—no per-task gradient descent. The project reproduces Drag-and-Drop LLMs (DnD) at a scale that fits one consumer GPU, then extends it along the axis where DnD is documented to be weakest: the small training-zoo regime. The deliverable is a clean, reproducible result with honest before/after numbers, plus a rigorous analysis of whether generated adapters truly resemble trained ones.

2 Background and prior art

Hypernetworks generate the parameters of a target network from a conditioning signal (Ha et al., 2016). Applied to LoRA (Hu et al., 2021), this yields adapters on demand. Two recent LLM-side systems anchor the design space:

- **Text-to-LoRA (T2L)** — conditions on a natural-language task description; a frozen text encoder plus learnable module/layer embeddings feed small MLPs that emit A and B per target. Trained by reconstruction or by end-to-end SFT; SFT generalized better.
- **Drag-and-Drop LLMs (DnD)** — conditions on a batch of unlabeled prompts; a text encoder feeds a cascaded hyper-convolutional decoder that emits the full LoRA set. Reported up to 12,000x lower overhead than fine-tuning and up to 30% gains over training LoRAs on unseen tasks, but these gains are sensitive to zoo diversity and collapse in the few-dataset regime.

On the diffusion side, HyperDreamBooth (Ruiz et al., 2023) and DiffLoRA (2024) generate personalized diffusion LoRAs from images. Doc-to-LoRA (Sakana, 2026) generalizes T2L to context-conditioning and demonstrates cross-modal transfer. Foundational to the weights-as-data view is Neural Network Diffusion / p-diff (Wang et al., 2024). VeRA (Kopiczko et al., 2023) is noted below as a considered-and-deferred adapter format.

3 Thesis and contributions

Thesis. Prompt-to-weights generation can be made to work in the compute-constrained, small-zoo regime that a single GPU imposes—the exact regime in which DnD reports it barely beats the baseline.

Contributions (two co-headlines).

- **Zoo augmentation for data-efficient generation.** Multiply a small adapter zoo using (a) training-trajectory checkpoints, which are free byproducts of zoo collection, and (b) interpolation between adapters, performed in a gauge-canonicalized weight space so it is well-defined. Measured as lift over an honest small-zoo baseline.
- **Generated-vs-real adapter analysis.** Rigorously test DnD's claim that generated parameters resemble trained ones—on weight-space distance, singular-value spectra, subspace overlap with the base weights they edit, and, critically, functional/behavioral equivalence (do they induce the same activation shifts, not merely similar-looking weights?).

4 Technical approach

4.1 Gauge canonicalization of LoRA targets

Raw LoRA is not gauge-free: infinitely many (A, B) pairs yield the same low-rank update BA, so both the regression target and any interpolation are ill-defined in A/B space. Fix this by canonicalizing the entire zoo via SVD: form the update BA, take its SVD, and store gauge-fixed factors (fold singular values consistently into one side). Every target then lives in one canonical gauge—this improves the reconstruction regression on its own and makes interpolation well-behaved. Always interpolate the update BA (or the canonical factors), never raw A and B.

4.2 Reproduction, scaled

Fork DnD; run on the smallest base (Qwen2.5-0.5B) using the authors' released zoo so collection is skipped for the reproduction itself. Apply canonicalized targets. Bank trajectory checkpoints from any zoo run from day one—the data you will want for augmentation is free but only if you save it.

4.3 QLoRA as an enabler, not a headline

QLoRA emits the same tensors as LoRA; it is a collection-time memory trick, not a separate generation mode. Its real jobs: collect a zoo on a larger base within 32 GB, and support one small robustness study (can the generator serve a 4-bit-deployed base as well as a full-precision one?). Kept in the toolbox; not the story.

5 Work plan (priorities)

Pri.	Item	Role in the project
1	Reproduce scaled DnD (+ optional conditioning enrichment)	Baseline + pipeline. A bare learnable token is not a step; enriching conditioning with a task descriptor (prompts vs prompts+description) is a legitimate small ablation.
2	QLoRA enabler + quantized-base robustness check	Extends reachable base size; minor deployment-relevant study. Not a headline.
3	Mech-interp / statistics: generated vs real adapters	Co-headline. Parallel analysis layer—startable the moment any adapters exist.
4	Trajectory + interpolation zoo augmentation	Co-headline and load-bearing novelty. Directly targets DnD's small-zoo weakness. Interpolate in canonicalized space.
5	Knowledge encoding on MLP layers (D2L-like)	Stretch / project two. Well-motivated (FFN-as-memory, ROME/MEMIT) but a separate task family and eval; space partly occupied by Doc-to-LoRA.

6 Four-week schedule with go/no-go checkpoints

Week	Focus	Go / no-go checkpoint
1	Environment (budget the Blackwell/CUDA day). Reproduce scaled DnD on released zoo; canonicalize targets; bank trajectory checkpoints.	Generated adapter beats the base model on a held-out task. If not, you are debugging the pipeline, not extending it.
2	Build a small own zoo (QLoRA for a bigger base if wanted). Establish the honest small-zoo baseline.	Does it generalize at all in your regime? A "no" here is your motivation section, not a failure.
3	Augmentation interventions (trajectory + interpolation), measured as lift over the Week-2 baseline. Core result.	Augmentation produces measurable lift over the un-augmented small-zoo baseline.
4	Generated-vs-real analysis; writeup and repo. Reserve the back half purely for the writeup—that is what gets read.	Clean repo + writeup with reproducible numbers.

Items 1's conditioning enrichment, 2's quantized-base study, and all of item 5 are explicitly future work so the scope stays a month.

7 Scope boundaries and deferred directions

VeRA as a generation target — deferred

VeRA offers a tiny, gauge-free target that is easier to generate and cleaner to interpret. But in a generation setting its usual wins (cheap training, small checkpoints, multi-tenant storage) all evaporate, and at inference it merges to the same update as LoRA with equal-or-worse quality—so the end user gains nothing. Its value was purely research-tractability. SVD canonicalization (4.1) recovers most of the gauge-freedom benefit for interp and interpolation while staying native to LoRA, so VeRA is dropped to keep one crisp contribution axis.

Extension to diffusion / cross-modal — deferred

Mechanically the hypernet is base-agnostic (it regresses weights with the base out of the loop), and DiT-based diffusion backbones resemble the LLM case. But the conditioning does not transfer: diffusion LoRAs are defined by images, not prompt batches, so porting DnD means swapping to an image encoder—which lands squarely in HyperDreamBooth / DiffLoRA's occupied territory. Evaluation also degrades from a clean scalar (accuracy) to fuzzy perceptual metrics, undermining the augmentation study's before/after numbers, and Doc-to-LoRA already claimed the cross-modal-generality point. Deferred; at most a one-figure appendix probe on an SD1.5 style-LoRA zoo after the LLM work is complete.

Weight-space latent diffusion over the zoo — promoted candidate

Rather than a deterministic prompt-to-weights regressor, train a small VAE or latent diffusion model over the canonicalized LoRA zoo and sample novel adapters, optionally task-conditioned via classifier-free guidance. The appeal is twofold: it is the richest interpretability story (is there a smooth, sample-able task manifold in LoRA space?), and—the reason it is attractive here—sampling is itself a zoo-augmentation engine, generating fresh adapters to enlarge the small training set. Adjacent to Neural Network Diffusion (p-diff) and DiffLoRA; the delta is conditional latent diffusion over LLM task adapters plus an explicit manifold analysis. Chief risk: a two-stage pipeline that can memorize the zoo rather than generalize, so memorization must be tested directly.

Cross-scale transfer via a shared trunk — promoted candidate

Structure the hypernet as a shared task-encoder trunk (conditioning to a task code) with a base-specific output decoder, then measure how much transfers when swapping bases within a family (e.g., Qwen2.5-0.5B to 1.5B). The trunk is genuinely base-agnostic and the per-base decoder is an implementation detail; the real question is empirical—does the same task occupy an analogous region of adapter space across models? Either outcome is a result: a positive one shows shared task structure across scales, a negative one shows adapters are highly base-idiosyncratic. DnD lists heterogeneous-architecture extension as future work, so the space is fairly open. Cost: two zoos, and the decoder must be re-headed per base since hidden dims and layer counts differ.

8 Environment and hardware notes

- **Blackwell (sm_120) toolchain.** Both Sakana repos pin flash-attention 2.6.3 (cu123/torch2.3) and DnD pins torch 2.5.1+cu124—none support Blackwell out of the box. Budget the first half-day to move to a torch 2.7+/CUDA 12.8 build and a newer flash-attn (or disable it).
- **Memory phases.** Zoo collection loads the base (QLoRA helps here); hypernet training in the reconstruction regime does not load the base at all; eval is inference-only. Keep the base frozen and in bf16; only the hypernet carries optimizer state.
- **Do not chase the papers' full runs.** T2L reports ~5 GPU-days on an H100; DnD assumes a multi-GPU box. On one 5090, run released checkpoints for demos and retrain the hypernet on small bases.

References

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